Environ. Eng. Res. 2025; 30(2): 240403
https://doi.org/10.4491/eer.2024.403

pISSN 1226-1025
eISSN 2005-968X

Research

Carbon emission prediction and reduction analysis of wastewater treatment plants based on hybrid machine learning models

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Received July 2, 2024 **Revised** August 24, 2024 **Accepted** September 20, 2024

ABSTRACT

Accurate accounting and prediction of carbon emissions from sewage treatment plants is the basis for exploring low-carbon sewage treatment plants and measures to reduce pollution and carbon emissions. This study proposes a hybrid prediction framework based on machine learning, which integrates multiple algorithms and has strong adaptability and generalization ability. The prediction framework uses Pearson correlation coefficient to select feature values, constructs a combined prediction model based on the selected features using support vector machine (SVR) and artificial neural network (ANN), and optimizes the SVR model parameters and structure using Gray Wolf Optimization (GWO) algorithm. The results show that the model has stronger prediction performance compared with other prediction models, with a mean absolute percentage error (MAPE) of 0.49% and an R2 of 0.9926. In addition, this study establishes six future development scenarios based on historical data trends and policy outlines, which provide recommendations for the development of carbon emission reduction measures for wastewater treatment plants. This study can provide a reference for exploring efficient carbon management and achieving carbon neutrality in wastewater treatment plants.

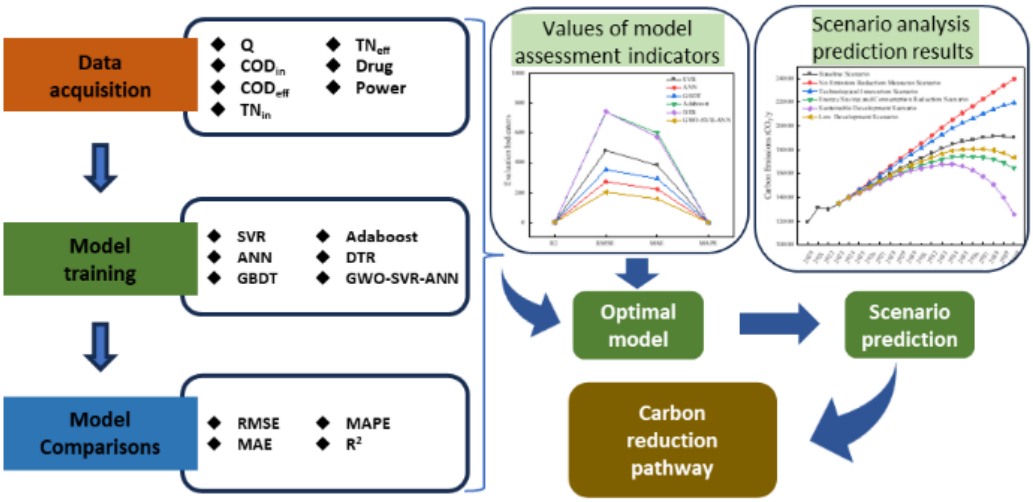
Keywords: Carbon accounting, Carbon emission, Carbon emission prediction, Carbon neutrality, Wastewater treatment

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Graphical Abstract



1. Introduction

The wastewater treatment sector is one of the ten highest emitters of greenhouse gases worldwide, accounting for around 2 to 3% of total carbon emissions. Moreover, its overall carbon footprint continues to rise annually. Thus, the industry must adopt carbon reduction measures to align with "dual carbon" objectives and foster synergies between pollution mitigation and carbon abatement [1]. While wastewater treatment significantly improves water ecosystem health by diminishing influent pollutants and achieving high-quality effluent, it also generates greenhouse gases and solid wastes, notably sludge, alongside considerable energy and resource consumption, all negatively impacting the environment [2]. Therefore, accurate accounting and prediction of carbon emissions from wastewater treatment processes is essential to optimize pollution and decarbonization of industry. These efforts can provide guidance on future emission trends and assist in the development of relevant energy efficiency and emission reduction strategies.

Analyzing and predicting GHG emissions from wastewater treatment plants based on accounting results can accelerate the decarbonization process of the wastewater treatment industry. Carbon emission prediction models have been widely used in the fields of wastewater treatment and carbon emission prediction. Specific methods including regression analysis, random forests, and neural networks have been widely used in real-world scenarios such as wastewater quality prediction, carbon emission estimation, and greenhouse gas emission impact assessment [3-8]. For example, Azeez et al. [9] developed a support vector regression (SVR) model to predict carbon emissions from automobiles, Zhu et al. [10] investigated and predicted the peak carbon emissions from the transportation sector in China using an SVR model and a scenario analysis model; furthermore, Chu and Zhao [11] used an enhanced PSO-SVR model to predict carbon emissions from buildings. Wodecka et al. [12] employed machine learning methods to systematically predict wastewater quality at a wastewater treatment plant inlet. Meanwhile, Vasilaki et al. [13] developed an SVR model to forecast N_2O concentration within both the anaerobic and aeration stages. Additionally, Alali et al. [14] conducted a comparative analysis of prediction accuracies among diverse machine learning models including ANN, SVR, KNN, etc., to estimate energy consumption in wastewater treatment.

The emission factor method is presently the most prevalent accounting approach, widely utilized at national and city scales due to its operational simplicity [15-20]. X. Zhao et al. estimated methane emissions from prefectural-level municipal wastewater treatment plants in China using this method [21], while Dan Wang et al. compiled an emission inventory for China's enterprise-level wastewater treatment plants spanning 2006-2019 [22]. Jiahui Han et al. employed a combination of the emission factor method and mass balance method to model on-site and off-site GHG accounting for paper mill WWTPs [23]. However, due to the lack of reliable emission factors, this method yields results with significant uncertainty. Presently, it primarily relies on emission factors provided by the IPCC. Nonetheless, research indicates that China's wastewater treatment plant emission factors are lower than those measured in European countries and by the IPCC [24], and variations exist in carbon emission factors among different provinces and treatment processes [25]. Hence, employing local carbon emission factors can effectively mitigate the accounting error associated with the emission factor method.

While carbon emission prediction models have found widespread application in various domains like construction and transportation, research in the wastewater treatment field remains limited. Although studies have been conducted to predict China's future wastewater emissions and reduction potential based on the IPCC Population Equivalent Approach model combined with scenario analysis, this method is only applicable to regional wastewater treatment plants [26]. However, this method is only applicable to regional wastewater treatment carbon emission prediction,

and its application to individual wastewater treatment plants is still limited. In contrast, the application of machine learning to predict carbon emissions, first, the overall prediction process is more streamlined, only need to input some of the values to get the carbon emissions in the coming years; second, the prediction includes fossil carbon emissions, which reduces the overall prediction error. This study develops a carbon accounting model using a literature review and local emission factor databases to assess carbon emissions from entire sewage treatment plants. Additionally, it utilizes treated water volume, effluent data, chemicals consumption, and energy consumption as input parameters to construct a hybrid prediction model for predicting carbon emissions from wastewater treatment plants. Finally, the prediction model is employed for scenario analysis to devise a rational emission reduction strategy based on the prediction outcomes.

2. Materials and Methods

2.1. Carbon Emission Accounting Method

This study focuses on accounting for carbon emissions during the operation and maintenance phase of the wastewater treatment plant, drawing primarily from the IPCC National Greenhouse Gas Guidelines and relevant literature on wastewater treatment plant carbon emission accounting [1, 27, 28]. The accounting boundary, illustrated in Fig. S1, encompasses direct emissions primarily from fossil sources, CH_4 emissions from anaerobic digestion, and N_2O emissions from denitrification. Only fossil carbon emissions are considered for CO_2 emissions in the biological treatment stage. Indirect emissions mainly account for carbon emissions from electricity and chemical consumption, purchased water sources, etc. Since the sludge is treated by thickening and transportation, only carbon emissions from thickening and dewatering are considered in the accounting. The main power-consuming equipment of sewage treatment plants are lifting pumps, grating machines, aeration equipment, sludge treatment equipment, etc. The chemicals are mainly flocculants, coagulants, phosphorus removers, disinfectants and so on. The specific calculation formula is as follows:

(1) The fossil carbon emissions in wastewater treatment plants are calculated as shown in Eq. (1), Eq. (2) and Eq. (3):

$$m_{CO_2} = Q \times MFCF$$

$$\left\{ \begin{array}{c} \left[1.1\left(B_{in}+B_{ex}-B_{eff}\right)\times\left(1.47-1.42\times\left(\frac{0.67}{1+K_d\cdot SRT}\right)\right)\right] \\ +(1.947HRT\cdot MLVSS\cdot K_d)-4.49\times \\ \left[\left(TN_{in}-TN_{eff}\right)-\left(B_{in}+B_{ex}-B_{eff}\right)\times\left(\frac{0.67}{1+K_d\cdot SRT}\right)\times 0.124\right] \end{array} \right\} \times 10^{-3} \quad (1)$$

$$\mathrm{MFCF} = \frac{FCF \times B_{in} + B_{ex}}{B_{in} + B_{ex}} \quad (2)$$

$$K_d = 0.05 \times 1.047^{T_b-20} \quad (3)$$

where m_{CO_2} is the fossil source emissions from wastewater treatment, in terms of emission equivalents produced by treating each cubic meter of wastewater, $kgCO_2$; Q is the daily volume of water treated in the wastewater treatment plant, m^3; $MFCF$ (Fossil carbon emission fraction) is the proportion of fossil sources discharged; SRT is the mean biosolids retention time, d; 1.947 is the yield of microorganisms corresponding to the consumption of synthetic cellular material per 1 kg of endogenous respiration, $kgCO_2/kg$ [29]; and HRT is the hydraulic retention time of the bioreactor, d; MLVSS is the mean mass concentration of volatile suspended solids in the mixed liquor of the bioreactor, mg/L; 4.49 is the mass fixed per unit mass of ammonia nitrogen nitrification, $kgCO_2/kg$; TN_{in} is the mass concentration of total nitrogen in the influent of the wastewater treatment plant, mg/L, TN_{eff} is the mass concentration of total nitrogen in the effluent of the wastewater treatment plant, mg/L, 0.124 is the mass ratio of the microorganisms ($C_5H_7O_2N$) body with N mass ratio; FCF is the proportion of fossil source organic matter in the influent water of the sewage treatment plant, taking the value of 10% [30]; 1.1 is the yield of microbial degradation of organic matter; B_{in} is the mass concentration of BOD in the influent water of the wastewater treatment plant, mg/L; B_{ex} is the additional carbon source artificially added during the operation, mg/L; B_{eff} is the effluent BOD of the sewage treatment plant, mg/L; 1.47 is the ratio of total BOD to BOD_5 of the influent water of the sewage treatment plant; 1.42 is the BOD_5 equivalence of the microbial cells, $kgCO_2$; 0.67 is the absolute yield coefficient, $kgCO_2/kg$; K_d is the decay coefficient, d; 0.05 is the reaction rate constant, d^{-1}; 1.047 is the temperature coefficient.

(2) The CH_4 in wastewater treatment plants are calculated as shown in Eq. (4):

$$m_{CH_4} = \frac{Q \times \left(B_{in} - B_{eff}\right)}{1000} \times EF_{CH_4} \quad (4)$$

where m_{CH_4} is the discharge of CH_4 in the sewage treatment plant, $kgCH_4$; Q is the daily volume of water treated in the sewage treatment plant, m^3; B_{in} is the sewage treatment plant influent BOD concentration, mg/L; B_{eff} is the sewage treatment plant effluent BOD concentration, mg/L; EF_{CH_4} is the methane emission factor, $kgCH_4/kgBOD$.

(3) The N_2O in wastewater treatment plants are calculated as shown in Eq. (5):

$$m_{N_2O} = \frac{Q \times \left(T_{in} - T_{eff}\right) \times EF_{N_2O}}{1000} \times C_{N_2O/N_2} \quad (5)$$

where m_{N_2O} is the discharge of N_2O in the sewage treatment plant, kgN_2O; Q is the daily volume of water treated in the sewage treatment plant, m^3; TN_{in} is the mass concentration of total nitrogen in the influent water of the sewage treatment plant, mg/L; TN_{eff} is the mass concentration of total nitrogen in the effluent water of the sewage treatment plant, mg/L; is the emission factor of N_2O, $kgN_2O/kgCOD$; C_{N_2O/N_2} is the ratio of the molecular mass of N_2O to that of N_2.

(4) Electricity consumption is the purchased electricity during the production and operation of the wastewater treatment plant, excluding the electricity consumption of office and living areas.

Table 1. Basic O&M data for wastewater treatment plants

Year	Q	BODin	CODin	TNin	BODeff	CODeff	TNeff
2012	38259	69	169	21.51	10.0	37	13.19
2013	33861	94	242	31.71	10.7	45	16.00
2014	34879	141	335	33.51	11.6	49	16.01
2015	37600	135	353	36.24	9.4	50	14.57
2016	39246	116	417	38.42	7.0	48	12.53
2017	38332	137	445	45.29	6.7	51	16.28
2018	42432	167	486	48.19	3.5	41	13.47
2019	52361	144	374	37.99	3.1	34	10.93
2020	68759	141	327	30.05	3.8	28	9.95
2021	73284	127	309	34.59	3.1	29	9.63
2022	82672	123	300	30.01	3.7	26	7.75

The calculation of carbon emissions from electricity consumption of wastewater treatment plants is shown in Eq. (6):

$$m_e = fe \times W \tag{6}$$

where m_e is the indirect emissions from sewage treatment power consumption, $kgCO_2$; fe is the carbon emission factor of power consumption, $kgCO_2/kwh$; W is the purchased electricity used for production and operation, kwh.

(5) Physical consumption is the coagulant, flocculant, carbon source, disinfectant and cleaning agent and other chemicals consumed during the production and operation of wastewater treatment plants. The calculation of carbon emissions form material consumption of wastewater treatment plants is shown in Eq. (7):

$$M_c = \sum_{g=1}^{m} \left(f_{c,g} \times M_{cg}\right) \tag{7}$$

where M_c is the emission equivalent of the consumption, $kgCO_2/kg$; $f_{c,g}$ is the emission factor for chemical g, $kgCO_2/kg$; M_{cg} is the amount of chemical g used, kg.

(6) The total carbon emissions from a wastewater treatment plant are calculated as shown in Eq. (8):

$$M_t = m_{CO_2} + m_{CH_4} \times 28 + m_{N_2O} \times 265 + m_e + M_c \tag{8}$$

where M_t is the total carbon emission from the operation and maintenance of the wastewater treatment plant, $kgCO_2$; 28 is the global warming potential (GWP) of CH_4, constant, $kgCO_2 / kgCH_4$; 265 is the global warming potential (GWP) of N_2O, constant, $kgCO_2/kgN_2O$.

2.2. Data Sources

2.2.1. Basic data for operation and maintenance of wastewater treatment plants

The wastewater treatment plant examined in this study is situated in the northeastern region of the Cixi Binhai Economic Development Zone, designed to process a daily treatment capacity of 10×10^4 m^3. Data utilized include the daily wastewater treatment volume, average daily pollutant concentrations entering and exiting the facility, daily electricity consumption, and daily chemical usage from 2012 to 2022. These data are sourced from the wastewater treatment plant's routine measurements and statistical records. Table 1 presents specific operation and maintenance (O&M) parameters for the WWTP.

2.2.2 Emission factor data

The N_2O and CH_4 emission factors of different treatment processes are summarized and summarized by checking relevant literature, as shown in Fig. S2 and Table S1, and the chemicals carbon emission factors are shown in Table S2, and the emission factors of China's East China Regional Power Grid for the year 2012-2022 are shown in Fig. S3.

2.3. Combined Forecasting Model

2.3.1. Grey wolf optimization algorithm (GWO)

Grey wolf optimization algorithm (GWO) is one of the meta-inspired optimization algorithms, which is based on the hunting behavior of grey wolves in Canidae [31]. As shown in Fig. S2(a) and (b), the gray wolf pack consists of α wolves, β wolves, γ wolves and ω wolves, α wolves dominate decision making about hunting location, sleeping time, etc., β wolves assist α wolves in decision making, β wolves take over the dominant position of α wolves when 2 wolves are in danger and unable to make decisions, γ wolves and ω wolves are located in the lowest rank of the pack, γ wolves provide services to α wolves and ω wolves, β wolves exist as scapegoats in the pack in order to succumb to the other wolves that are dominant. The GWO optimization algorithm firstly hierarchizes the wolves into a social hierarchy and then stalks and surrounds its prey, the mathematical model of this behavior could be represented as Eq. (9), Eq. (10), Eq. (11), Eq. (12):

$$\vec{\mathrm{D}} = \vec{\mathrm{C}} \cdot \overrightarrow{X_p}(t) - \vec{X}(t) \tag{9}$$

$$\vec{X}(t+1) = \overrightarrow{X_p}(t) - \vec{A} \cdot \vec{D} \tag{10}$$

$$\vec{\mathrm{A}} = 2\vec{\mathrm{a}} \cdot \overrightarrow{r_1} - \vec{a} \tag{11}$$

$$\vec{\mathrm{C}} = 2\overrightarrow{r_2} \tag{12}$$

where t is the current iteration number, $\vec{A}$ and $\vec{\mathrm{C}}$ are the coefficient vectors, $\overrightarrow{X_p}$ denotes the position vector of the prey, $\vec{X}(t)$ denotes

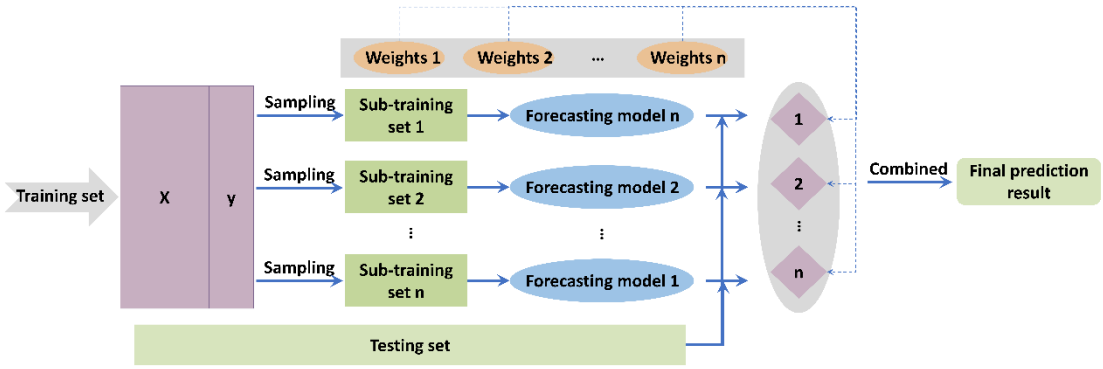


Fig. 1. Combined modelling framework.

the current position vector of the gray wolf, and a decrease linearly from 2 to 0 throughout the iteration process and $\vec{r_1}$, $\vec{r_2}$ are random vectors between [0,1].

Gray wolves have the ability to identify the location of potential prey (optimal solution), and the search process is mainly accomplished by the guidance of α wolves, β wolves and γ wolves. However, the solution space characteristics of many problems are unknown, and the gray wolf is unable to determine the precise location of the prey (optimal solution). In order to simulate the search behavior of the gray wolf (candidate solution), it is assumed that the α wolves, β wolves, γ wolves have a strong ability to identify the location of the potential prey. Therefore, during each iteration, the best three gray wolves (α wolves, β wolves, γ wolves) in the current population are retained, and then the locations of the other search agents are updated based on their location information. The mathematical model of this behavior can be expressed as Eq. (13), Eq. (14), Eq. (15):

$$\vec{D}_\alpha = \vec{C_1} \cdot \vec{X_\alpha} - \vec{X}, \vec{D}_\beta = \vec{C_2} \cdot \vec{X_\beta} - \vec{X}, \vec{D}_\gamma = \vec{C_3} \cdot \vec{X_\gamma} - \vec{X} \tag{13}$$

$$\vec{X_1} = \vec{X_\alpha} - \vec{A_1} \cdot \vec{D}_\alpha, \vec{X_2} = \vec{X_\beta} - \vec{A_2} \cdot \vec{D}_\beta, \vec{X_3} = \vec{X_\gamma} - \vec{A_3} \cdot \vec{D}_\gamma \tag{14}$$

$$\vec{X}(t+1) = \frac{\vec{X_1} + \vec{X_2} + \vec{X_3}}{3} \tag{15}$$

2.3.2. Support vector machine (SVR)

2.3.3. Artificial neural network (ANN)

ANN model consists of three parts: input, hidden and output layers, Neural Network (NN) processes the input data and indicates the significance relationship between the input variables and the output mechanism. It can learn from the original dataset and test the generated network with a new dataset, further correct and strengthen the internal logical relationships for the training model, and finally validate the model with a validation dataset. The specific model explanation is shown in Text S1.

Compared with traditional prediction models, ANN has superior learning ability in multi-input data processing, and can quickly sort out the hidden internal logic of the input and output layers, and this internal structure enables it to effectively manage and interpret the original time series data. Meanwhile, ANN has superior ability in mining the nonlinear features of total carbon emissions from sewage treatment plants, and this ability significantly improves the accuracy of carbon emission prediction.

2.3.4. Combined prediction model

Different prediction models have their strengths and weaknesses, and to maximize the strengths and avoid the weaknesses of the combination model is introduced. The combined prediction model integrates the prediction results of multiple models according to different weights, and the combined new model combines the prediction advantages of the sub-models to effectively improve the prediction performance of the model. The framework of the method is shown in Fig. 1. The framework first divides the dataset into a training set and a test set, trains multiple predictive models based on the feature or attribute X and the target variable Y of the data, and assigns different weights to the models, so that the optimal combination of weights of the model evaluation indexes is the combination model that we want, and validates the combination model using the test set.

The calculation method of the combined model is shown in Eq. (16), Eq. (17), Eq. (18): for a given model j with input X_i, the output Y_{ij} is obtained. Each model output value is assigned a weight α_j. The predicted value of the combined model is the combination of the output results Y_{ij} and their corresponding weights α_j, represented as W_i. This process determines the optimal weight for the model evaluation indices, resulting in the final combined model.

$$W_i = \sum_{j=1}^{k} \alpha_j \times Y_{ij}, i = 1,2,\cdots,k \tag{16}$$

$$\sum_{j=1}^{k} \alpha_j = 1 \tag{17}$$

$$\text{MinRMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(W_i - y_i)^2} \tag{18}$$

2.3.5. Model evaluation indicators

Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE) and Goodness of Fit (R^2)

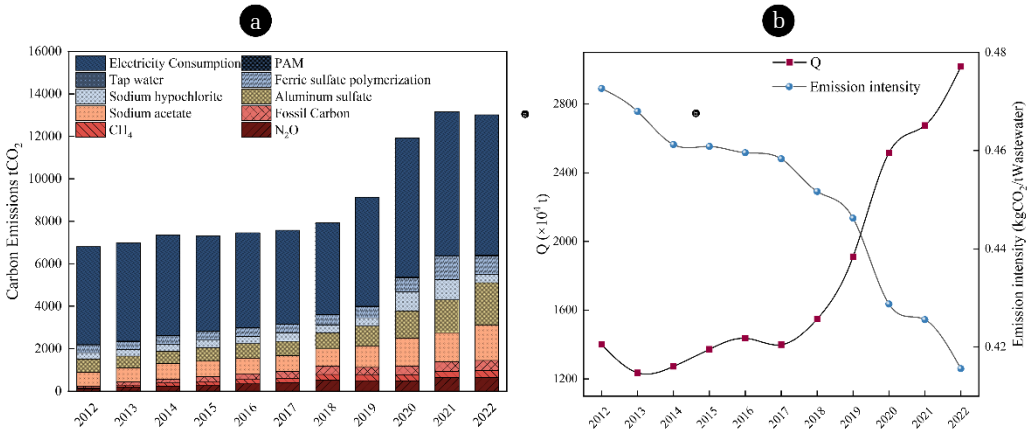


Fig. 2. (a) Total carbon emissions from 2012 to 2022. (b) Carbon emission intensity and water treatment capacity from 2012 to 2022.

were chosen as the indicators for judging the final prediction results, and the relevant Eq. (19), Eq. (20), Eq. (21), Eq. (22) are as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - y_{pre}| \tag{19}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - y_{pre})^2} \tag{20}$$

$$\mathrm{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n} \left|\frac{y_i - y_{pre}}{y_i}\right| \tag{21}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - y_{pre})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{22}$$

where y_i is the actual water quality monitoring value, y_{pre} is the simulated prediction value, $\bar{y}$ is the average value of measured water quality, n is the number of samples. The value range of MAE, RMSE, MAPE is [0, +∞], and the value range of [0, 1]. A smaller value for MAE, RMSE, and MAPE, as well as a value closer to 1 for R^2, indicates better prediction performance [32].

3. Results and Discussion

3.1. Carbon Accounting for Sewage Treatment

Fig. 2(a), (b) show plots of total carbon emissions and carbon emission intensity versus volume of water treated for WWTPs for the period 2012 to 2022, respectively, which were calculated using the improved carbon accounting methodology and carbon emission factor data. The overall trend reveals a continuous increase in total carbon emissions over the years, with indirect carbon emissions emerging as the primary contributor to the WWTP's total carbon footprint. Particularly noteworthy is the significant contribution of carbon emissions from electricity consumption. However, direct carbon emissions account for only a small portion of total carbon emissions, which is attributed to the process application of inverted AAO. In contrast to the conventional AAO process, the inverted AAO process reverses the anaerobic and anoxic positions. This allows the microorganisms to consume a significant quantity of the primary carbon source in the anaerobic zone. When the nitrifying solution in the aerobic zone flows back to the anoxic zone, the lack of carbon source in the traditional AAO process results in insufficient denitrification and the release of more N_2O [33]. This results in the lowest N_2O and CH_4 emission factors of the inverted AAO process. Furthermore, the direct carbon emissions from this plant will only account for 12% of the total carbon emissions in 2022. Concurrently, the influent of this wastewater treatment plant is a mixture of domestic and industrial wastewater, which contains a high concentration of nitrogen (N), phosphorus (P), and industrial carbon. This results in a higher proportion of N_2O and fossil carbon emissions than CH_4.

3.2. Analysis of Influencing Factors

In machine learning, feature values usually refer to the important attributes or metrics of each feature in the data. They help to identify and quantify the most important information in the data, thus improving the efficiency and performance of the model. Pearson's correlation coefficient based on covariance analysis can effectively check the correlation and extract features without directly affecting the model analysis [34]. The heat map of correlation analysis shown in Fig. 3 shows that there is a significant correlation between the total carbon emissions and the volume of treated water, effluent COD concentration, effluent TN concentration, and the consumption of various types of chemicals. In order to improve the prediction performance and efficiency of the model, the treated water volume, electricity consumption, effluent COD concentration, effluent TN concentration and total chemicals consumption were chosen as inputs to the model, and the total amount of carbon emission was chosen as the output of the model.

In this study, there may be multicollinearity between the influencing factors, which in turn may lead to feature redundancy or model overfitting. Therefore, variance inflation factor (VIF) was introduced to assess multicollinearity. In the regression model, the variance inflation factor (VIF) provides a measure of the degree of covariance; if VIF < 5, there is essentially no covariance, whereas a VIF above 10 should be considered to be covariant [35]. According to the results in Table 2, the VIF values between most of the

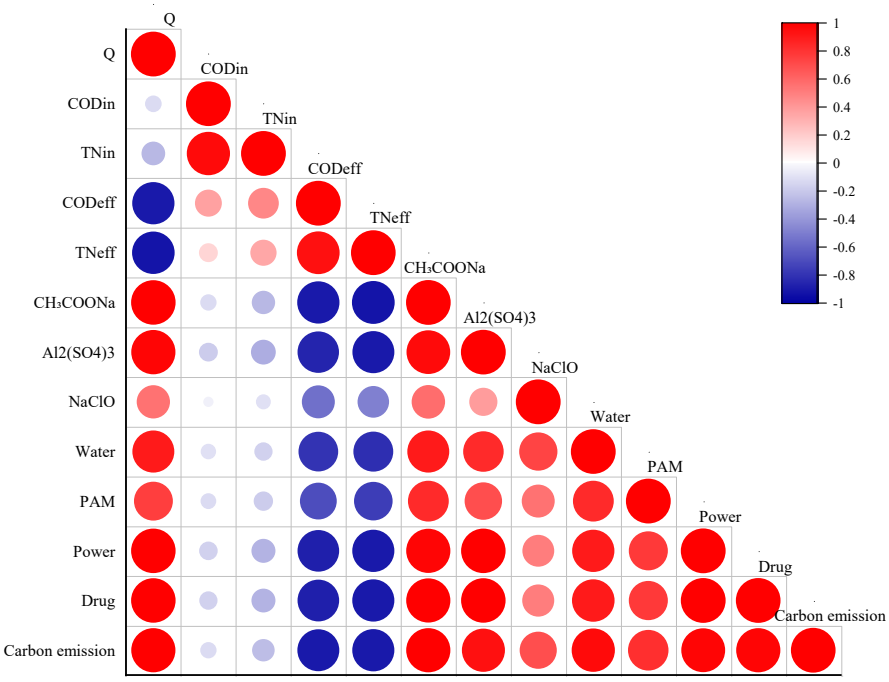


Fig. 3. Heat map for correlation analysis.

Table 2. VIF test results

Influencing factors	Significance	Covariance statistics	
		Tolerance	VIF
CODeff	0.000	0.500	1.999
TNeff	0.000	0.885	1.130
Electricity consumption	0.000	0.104	9.571
chemicals consumption	0.000	0.322	3.106
Q	0.000	0.117	8.570

influencing factors are below 10, indicating that there is basically no significant covariance between these factors. This indicates that in the regression model under consideration, the characteristic variables are mostly independent and do not interfere too much with the explanatory and predictive power of the model.

However, while the VIF values of most of the variables meet the criteria for no significant covariance, vigilance is still required for individual variables with VIF values close to 10. These variables may have some degree of slight covariance that may affect the stability and explanatory power of the model. Therefore, Principal Component Analysis (PCA) was applied in further analysis to further reduce redundancy and improve the predictive accuracy of the model.

3.3. Model Construction and Evaluation

Normalizing the data before model construction mitigates the impact of varying data scales on model performance and enhances its accuracy and robustness. Common normalization techniques include Min-Max normalization, Z-Score normalization, decimal scaling normalization, sigmoid normalization, and Robust-Scaler normalization. In this study, Min-Max normalization was employed, with the calculation being performed in accordance with the Eq. (23):

$$B' = \frac{B - B_{Min}}{B_{Max} - B_{Min}} \tag{23}$$

where B' is the normalized data, B is the original data, B_{Min} is the minimum value of the original data column and B_{Max} is the maximum value of the original data column.

In this paper, the daily cumulative data of this WWTP for the years 2012-2022 were selected to train the model, and the model was validated with the daily average data for the years 2012-2022 to assess the accuracy of the model predictions. The daily cumulative data is partitioned into a training set comprising the initial 80% and a test set consisting of the remaining 20%. In order to enhance the prediction accuracy of the Support Vector Regression (SVR) model, the Grey Wolf Optimization algorithm (GWO) is employed for parameter optimization, including the selection of the optimal kernel function. Through a systematic comparison of model scores across various parameter combinations, the optimal parameter configuration is determined. Notably, the linear kernel function emerges as the most effective choice, with

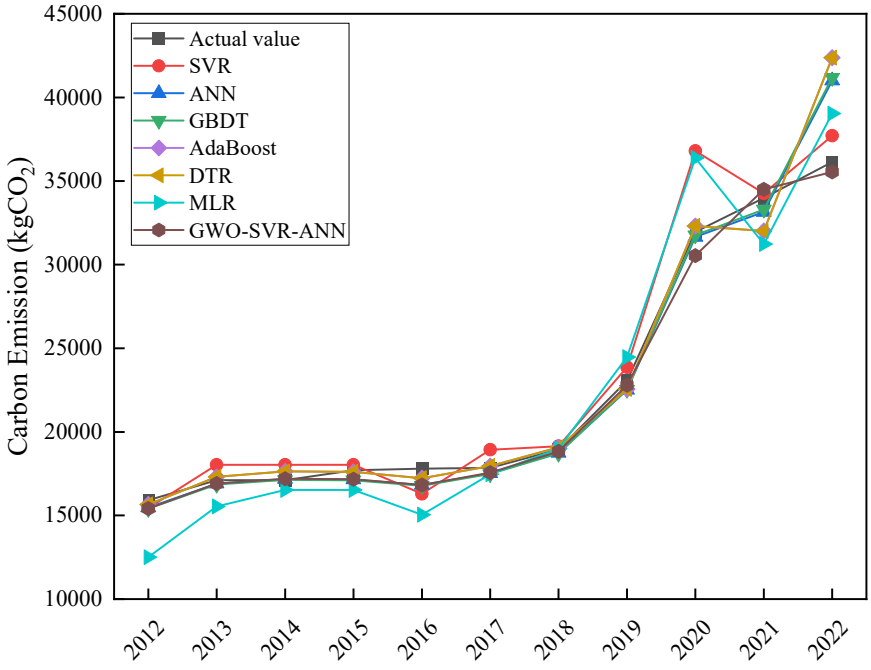


Fig. 4. Model Prediction Results.

Table 3. Values of model assessment indicators

	Model	SVR	ANN	GBDT	AdaBoost	DTR	MLR	GWO-SVR-ANN
test	R2	0.9590	0.9868	0.9779	0.9038	0.9027	0.8888	0.9926
	RMSE	483.25	274.32	354.85	739.90	744.30	832.27	204.87
	MAE	383.49	223.42	293.94	601.15	572.85	754.98	158.70
	MAPE	1.19%	0.70%	0.91%	1.87%	1.76%	5.04%	0.49%
train	R2	0.9577	0.9818	0.9997	0.9235	1.0000	0.8791	0.9934
	RMSE	518.32	339.94	43.94	697.35	0.00	852.48	204.98
	MAE	402.75	237.45	33.63	601.12	0.00	803.67	154.47
	MAPE	1.23%	0.73%	0.10%	1.85%	0.00%	6.10%	0.47%

optimal parameter values identified as C = 70816 and gamma = 48739. On the other hand, the Artificial Neural Network (ANN) model is structured with a two-layer architecture. The first layer incorporates a hidden layer comprising 16 neurons, while the second layer features five hidden layers, each housing 32 neurons. This configuration has been selected in order to facilitate robust learning and prediction within the ANN framework.

To enhance the overall predictive accuracy of the model, a hybrid approach combining the GWO-SVR and ANN models was employed, with weights assigned to each model to optimize performance. The optimal weight ratio was determined by minimizing the Root Mean Square Error (RMSE), thereby leveraging the predictive strengths of both sub-models to improve overall performance. The calculated optimal weight distribution assigns a weight of 0.404 to the ANN model and 0.596 to the SVR model, achieving the highest predictive performance.

To evaluate the effectiveness of the combined model, several comparative models were introduced, and their prediction results are summarized in Table 3. Notably, the combined model yields a remarkable R-squared (R^2) value of 0.9926, signifying a substantial improvement in prediction accuracy compared to standalone SVR and ANN models. Additionally, error metrics including RMSE, MAE, and MAPE demonstrate significant reductions, underscoring the superior performance of the combined model. The fitting results for the model test set are shown in Fig. S6. The fitting results for the validation set are shown in Fig. 4.

The GWO algorithm, functioning as a heuristic optimization technique, plays a pivotal role in aiding both SVR models to attain more optimal parameter configurations during training. Furthermore, compared to individual models, the combined approach exhibits enhanced robustness, mitigating the risk of overfitting to some extent.

Although Gradient Boosting Decision Trees (GBDT) and Decision Tree Regression (DTR) outperform SVR and ANN models

within the training dataset, their susceptibility to overfitting poses a limitation due to the constrained number of training samples. Moreover, these tree-based models are unable to simultaneously capture both linear and nonlinear relationships within the data, thus imposing constraints on their predictive capabilities.

3.4. Scenario Prediction

3.4.1. Future rate of change settings for different indicators

The parameters defining three development scenarios (low, medium, and high speed) for the seven development indicators underwent adjustments based on historical data from the wastewater treatment plant, coupled with the "Outline of the Fourteenth Five-Year Plan for the Development of the National Economy and Society of Ningbo Municipality and the Visionary Targets for the Year 2035" [36], alongside the "Synergetic Control of Environmental Pollution and Carbon Emissions Implementation Program" [37].

The medium-speed development scenario mirrors the average rate of change observed over the preceding five years. Conversely, the high-speed and low-speed development scenarios were formulated and fine-tuned in alignment with the national green low-carbon policy and the objectives outlined in Ningbo's ecological planning and construction initiatives. Detailed scenario change rates are delineated in Table S3.

We use pollutant concentration as an example to illustrate how to set parameters for each scenario. According to the statistical data of this wastewater treatment plant, the average annual growth rate of pollutant emission concentration is -9.3% during 2018-2022, so we set this value as the medium development rate of pollutant emission concentration. The Outline shows that the average annual growth rate of chemical oxygen demand and ammonia nitrogen emissions of major pollutants in Ningbo during 2015-2020 is -6.66%. We set this value as the low development rate of pollutant emission concentration, and the high development rate is adjusted accordingly based on the medium and low development rates. The clean energy substitution rate is set according to the plant's existing photovoltaic electricity utilization rate and future development plan.

3.4.2. Scenario design

To project the carbon emissions of the wastewater treatment plant (WWTP) over the next 20 years, this study established six distinct scenarios, outlined in Table S4. These scenarios aim to assess the carbon emissions and abatement potential of the WWTP under various developmental contexts, to evaluate the impact of different abatement strategies on total carbon emissions. By integrating considerations such as clean energy substitution, biological carbon sequestration, methane recovery for power generation, and other abatement techniques to fulfil the sustainable development objectives of WWTPs, this research offers valuable insights for other WWTPs in Ningbo. These insights aid in formulating carbon abatement measures and sustainable development strategies.

1) Baseline Scenario: The Baseline Scenario functions as a control group for the other scenarios. It assumes the continuation of the existing operational mode without alterations, with annual rates of change for different indicators based on reasonable extrapolations of historical data, all set to a medium level. Additionally, the projected outcomes of this scenario reflect the anticipated trend of carbon emissions from the wastewater treatment plant in the future.

2) No Emission Reduction Measures Scenario: No Abatement Measures Scenario: No Emission Reduction Measures Scenario is the control group against which other abatement scenarios are compared. It holds the rate of change in treated water volume, pollutant concentrations, energy consumption, and chemical consumption at moderate levels. The scenario's unchanged rates of clean energy substitution, methane recovery for electricity generation, and biocarbon sequestration mean that the scenario assumes that the WWTP maintains its current status without any additional abatement measures. The scenario assumes that the WWTP maintains its original treatment pattern and does not implement any emission reduction measures.

3) Technological Innovation Scenario: The scenario sets energy and chemicals consumption at a low growth rate, energy substitution rate and biological carbon sequestration rate at a high growth rate, and other indicators at a medium growth rate. Through the introduction of new management policies and technological paths to reduce energy and chemicals consumption in the treatment process, keep the growth rate of energy and chemicals consumption low in order to achieve the efficient use of resources, and at the same time increase the rate of production and proportion of clean energy in the plant, adjust the energy structure of the plant, and reduce the direct carbon emissions of the plant by combining with the advanced biological carbon sequestration technology, so as to promote the simultaneous development of efficient governance and low-carbon construction.

4) Energy Saving and Consumption Reduction Scenario: Energy conservation and consumption reduction serve as proactive measures for source control, fostering synergies between pollution mitigation and carbon reduction. This is achieved through enhancing resource and energy conservation and efficiency, and expediting the transition towards industrial structures, production methods, and lifestyles conducive to both pollution and carbon reduction. The scenario prioritizes water and electricity conservation, aiming to maintain a low growth rate in treated water volume and energy consumption. However, water resource reuse may elevate pollutant concentrations in water bodies. Hence, the rate of pollutant concentration reduction is set low, while other indicators experience moderate growth rates.

5) Sustainable Development Scenario: The sustainable development scenario is defined as a coordinated development of the economy, society and the environment. In this scenario, the wastewater treatment plant introduces new treatment technologies while achieving energy savings and consumption reduction. Accordingly, the proportion of clean energy substitution, the rate of methane recovery for power generation and the rate of biological carbon sequestration are set to a high-growth level in order to reduce the emission of pollutants to the maximum extent possible.

6) Low Development Scenario: The Low Development Scenario is marked by the plant's conservative approach to future planning and the delayed updating of treatment equipment and technology. Consequently, there is a pronounced upward trend in energy and chemical consumption. Moreover, inadequate capital investment

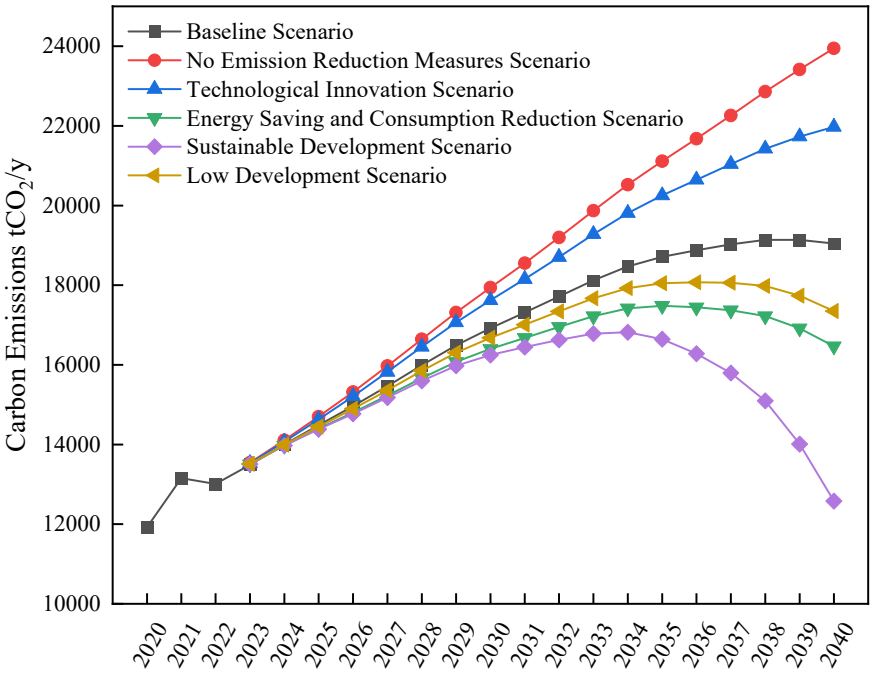


Fig. 5. Scenario analysis prediction results.

has resulted in a minimal increase in the proportion of clean energy substitution, methane recovery power generation rate, and biological carbon sequestration rate.

The carbon emission projections under different scenarios are reflected by the annual rate of change of the inputs (effluent COD, effluent TN, Amount of treated water, electricity consumption, chemicals consumption) under the scenario, and the inputs for each year are adjusted according to the rates set for different scenarios, and the carbon emission projections for each year are calculated on the basis of the abovementioned prediction model. At the same time, the final carbon emission forecast is calculated by combining the emission reduction programs under different scenarios.

3.4.3. Scenario prediction analysis

The results depicted in Fig. 5 illustrate the scenario analysis projections. Findings indicate that under the no-abatement- measures scenario and low-development scenario, the WWTP will not achieve carbon peak before 2040. Conversely, the baseline scenario, energy-saving scenario, technological innovation scenario, and sustainable development scenario forecast carbon peak occurrences in 2039, 2036, 2035, and 2034, respectively. Their respective peaks are projected at 19,138 t, 18,073 t, 17,484 t, and 16,819 t, approximately 1.37 times that of 2022. Notably, the no abatement measures scenario and low development scenario lack sustainable development potential. Furthermore, neither the baseline scenario nor the energy saving, and consumption reduction scenario can attain carbon peak by 2035. To meet the carbon peak target for water pollution prevention and control by 2030, it's imperative to bolster the proportion of clean energy substitution, methane recycling for power generation, and the rate of biological carbon sequestration as per the technological innovation scenario and sustainable development scenario. Additionally, efforts should be intensified to fortify prevention and control measures at the source, restrain high-pollution and high-emission industries and projects, promote water-saving and low-carbon lifestyles, and prioritize the adoption of advanced, low-energy-consuming process technologies and equipment.

3.5. Carbon Reduction Pathway Analysis

3.5.1. Energy conservation and consumption reduction

Carbon emissions from chemicals consumption and electricity consumption are the main sources of total carbon emissions from wastewater treatment plants, with chemicals consumption accounting for 35.4% of the total carbon emissions from wastewater treatment plants, and the total carbon emissions from chemicals consumption during a half-year period can be up to 2811 tCO_2eq. Sodium acetate and other additional carbon sources can be used to reduce the dosage by means of carbon cycling and carbon recycling, thus improving the sustainability of wastewater treatment plants. Additional carbon sources, disinfectants, flocculants, and other pharmaceuticals should be purchased with priority given to those that meet sustainability criteria to reduce indirect greenhouse gas emissions, and recycling to reduce the carbon emissions associated with the pharmaceutical production process. The indirect carbon emissions from the power consumption of this wastewater treatment plant accounted for 51% of the total carbon emissions, and the average power consumption reached 0.36 kWh/m^3. At present, the average power consumption of China's urban wastewater treatment plants is 0.29 kWh/m^3 and the power consumption of more than 82% of the wastewater treatment plants is not less than 0.440 kWh/m^3, so there is still a great potential for energy

saving in this wastewater treatment plant. Electricity consumption in municipal wastewater treatment plants mainly occurs in the three parts: the sewage lifting system, the oxygen supply system of the secondary biochemical treatment and the sludge treatment system, and the reduction of electricity consumption can be achieved from the perspectives of both power saving and clean energy substitution. It has been found that electricity consumption can be reduced by 7 - 9% by adjusting the sludge age and dissolved oxygen concentration in water, while the use of energy-saving equipment can further reduce energy consumption [38]. Some emerging reaction devices such as aerated membrane bioreactors, which can significantly increase the oxygen transfer rate, and precision aeration systems can reduce the energy consumption of the aeration process. Currently most of the electricity used is coal-fired power generation, increasing the use of renewable energy such as wind power, nuclear power, photovoltaic power can also reduce the indirect carbon emissions of sewage treatment plants, can be in the sand sedimentation tanks, secondary sedimentation tanks and unused land in the plant to increase the proportion of photovoltaic power to reduce the total carbon emissions of the whole plant.

3.5.2. Energy recycling

There is a large space for energy recovery in the wastewater treatment process, and the use of water source heat pumps to recover part of the secondary wastewater heat energy can reduce the energy consumption of the plant, thus reducing the total carbon emissions of the plant. Furthermore, there are four main sludge treatment technologies: mechanical dewatering, thermal drying, anaerobic digestion, aerobic fermentation [39], a large amount of biogas will be produced in the process of anaerobic digestion of sludge, and upgrading the discharged biogas to bio-methane, and reducing the energy consumption of the plant through biogas cogeneration can also reduce the total carbon emissions of the plant to a certain degree. For the sludge after mechanical dewatering and thermal drying treatment, sludge incineration can be used to generate electricity to supply energy for the plant.

3.5.3. Optimization of the treatment process

Efficient wastewater treatment process can improve the effluent quality and reduce carbon emission at the same time, for AAO wastewater treatment process, low DO concentration promotes the production of N_2O and high DO concentration inhibits the production of N_2O, meanwhile, the increase of aeration will increase the production of N_2O and vaporization, in addition, the low temperature will reduce the enzyme activity, which affects the nitrification and denitrification reactions, so maintaining complete nitrification without inhibiting the complete heterotrophic denitrification process by controlling DO and temperature at a reasonable level and controlling a certain amount of aeration is an effective way to reduce N_2O production [40]. For direct emission of the greenhouse gas methane, co-digestion techniques can be used to increase biogas and methane production and reduce the operating costs and energy requirements of the plant [41].

3.5.4. Enhancing carbon sinks

Biomass sequestration and energy recovery from biogas conversion are the two main strategies for carbon sequestration and sinks [42]. in addition to the use of microalgae-bacterial microbiomes to capture carbon dioxide during biodegradation of organic compounds to reduce direct greenhouse gas emissions [43], and the use of technologies such as microbial electrochemical technology (MET), photocatalytic, and electrochemistry to convert carbon dioxide into utilizable chemicals, such as formic acid, methanol, and methane, to enhance carbon sinks [44]. However, these technologies may have a negative impact on total carbon emissions, such as capture technologies that rely on high-carbon electricity, which may increase overall carbon emissions; material use and resource consumption also affect the environment, low technical efficiency or improper operation and maintenance may also lead to lower actual carbon reduction results than expected; some carbon sequestration projects, such as afforestation, may be affected by climate change. Therefore, a comprehensive assessment of these factors is very important for optimizing emission reduction strategies.

4. Conclusions

In conclusion, this study proposes a prediction framework that applies to carbon emissions from wastewater treatment plants with a high degree of prediction accuracy. The prediction framework initially located certain parameter values or coefficients provided in the IPCC and then combines them with a literature analysis to collate and update the carbon emission factors. It replaces the previous international data with the baseline emission factors for regional power grids issued by the Ministry of Ecology and Environment of China in different years and updates the GWP values of CH_4 and N_2O to the latest public data of IPCC 2019. Concurrently, the model incorporates fossil carbon into the accounting process, thereby resolving the issue of underestimation of the proportion of fossil carbon emissions in the traditional emission factor accounting method. Subsequently, the SVR and ANN prediction models were developed by combining covariance analysis and correlation analysis to extract features, and the GWO algorithm was used to further optimize the model parameters to improve the model's prediction performance. Finally, by minimizing the root mean square error (RMSE) and combining the two models according to the optimal weights, the coefficient of determination (R^2) of the model reaches 0.9926, and the mean absolute percentage error (MAPE) reaches 0.49%, which demonstrates superior prediction performance. This combined model effectively integrates the strengths of traditional machine learning and deep learning, capitalizing on the respective advantages of both. The introduction of nonlinear feature extraction capability, derived from deep learning, in conjunction with the generalization and interpretability of machine learning, enables the model to more accurately capture the complex nonlinear relationships between the data, thereby improving the prediction accuracy and robustness of the model.

Concurrently, this study employs scenario analysis, integrating the composite prediction model with pertinent policy frameworks to assess the carbon emissions of the wastewater treatment plant and propose targeted emission reduction strategies. Findings indicate that no emission reduction measures scenario and low devel-

opment scenario lack sustainable potential, while baseline scenario and energy saving, and Consumption Reduction Scenario fall short of achieving carbon neutrality by 2035. realizing the 2030 target for carbon neutrality in water pollution prevention and control necessitates augmenting the share of clean energy substitution, methane-recycling power generation, and biological carbon sequestration rates, as outlined in the technological innovation scenario and sustainable development scenario. Moreover, robust source prevention and control measures are imperative, alongside efforts to curtail high-pollution industries, promote water-saving and low-carbon lifestyles, and prioritize the adoption of advanced, low-energy process technologies and equipment. This study lays groundwork for optimizing wastewater treatment processes towards "zero carbon" and serves as a blueprint for effective carbon management in wastewater treatment plants, facilitating the journey towards carbon neutrality.

Acknowledgments

We gratefully acknowledge supports by the Baima Lake Laboratory Joint Funds of the Zhejiang Provincial Natural Science Foundation of China under Grant No. LBMHY24E060014.

Conflict-of-Interest Statement

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of.

Author Contributions

F.L. (M.D. student) led the Conceptualization, methodology, formal analysis and investigation and writing original draft preparation. N.D. (Professor) contributed to formal analysis and investigation, writing review & editing, funding acquisition and supervision. G.Z. (Tutor) participated in manuscript review & editing. J.X. (Professor) handled writing review & editing and supervision.

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